AEGNER BILLIK

TECHNICAL PORTFOLIO & ARCHITECTURAL CASE STUDIES

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PRODUCTION WEB & ENTERPRISE ARCHITECTURE

Koperasi UMB — Enterprise ERP & Offline POS (Bermadani) *2024 – 2026*

Laravel 11, React.js, PostgreSQL, Capacitor (Android APK), IndexedDB (Dexie.js), Bluetooth SPP | Live: bermadaniumbandung.id

- Architected an integrated dual-platform enterprise ERP and retail POS system serving university staff, faculty members, and campus minimarket operations.
- Engineered offline-first cashier capabilities using client-side IndexedDB caching; enabled uninterrupted checkout transactions during network outages with automated asynchronous background syncing upon reconnection.
- Implemented native Android tablet printing protocol via Capacitor and Java Bluetooth bridge, allowing high-throughput receipt generation on 58mm/80mm thermal printers.
- Constructed backend modules for multi-tiered member savings, automated payroll deduction loans, consignment supplier margins, and automated General Ledger journal entries.

Academic Accreditation Portal & Study Program CMS *2024 – 2025*

Next.js 14 (App Router), TypeScript, Tailwind CSS, PostgreSQL, RESTful API | Live: tekpang.umbandung.ac.id

- Developed an official university study program portal and multi-tenant Content Management System (CMS) supporting national BAN-PT accreditation compliance for Food Technology and Biotechnology.
- Leveraged Next.js Server Components and Incremental Static Regeneration (ISR) to achieve sub-second page loads and 100% SEO scores across academic documentation portals.
- Designed relational PostgreSQL database schema with role-based access control (RBAC) separating administrative faculty staff, lecturers, and student document access.

Automated Admissions Document OCR Extraction (PMB) *2024 – 2026*

Next.js, Python, OpenCV, Tesseract OCR, RegEx, FastAPI | Live Web App: ocr-pmb-umbandung.vercel.app

- Built an automated computer vision pipeline to ingest and parse photographic images of national ID cards (e-KTP) for prospective student admission forms.
- Applied OpenCV preprocessing filters (adaptive thresholding, noise reduction, and perspective correction) paired with Tesseract OCR and regular expressions to parse identity fields (NIK, Full Name, DOB, Address).
- Eliminated manual applicant typographical errors and accelerated application verification time by over 70% during peak registration waves.

HARDWARE, IOT & APPLIED MACHINE LEARNING SYSTEMS

Real-Time Factory Quality Control & Telemetry System *2024*

ESP32 Microcontroller, RFID (RC522), WebSockets, Redis Pub/Sub, React.js, C/C++ | Repo: github.com/BroAegg/Monitoring-QC-RFID

- Engineered an industrial hardware-software tracking system deploying RFID sensors and ESP32 microcontrollers across manufacturing quality inspection stations.
- Established bidirectional WebSocket telemetry via Redis pub/sub, instantly relaying card-tap technician events to live React monitoring dashboards with latency under 12ms.
- Eliminated paper-based manual QC logs, ensuring complete traceability and automated timestamp logging across manufacturing workflows.

Deep Learning Audio Genre Spectrogram Classifier *2024*

PyTorch, Librosa DSP, Convolutional Neural Networks (CNN), Streamlit, Python | Repo: github.com/BroAegg/Music_Genre_Classification

- Engineered an audio Digital Signal Processing (DSP) pipeline converting raw WAV audio streams into Mel-Frequency Cepstral Coefficients (MFCC) and Mel-spectrogram images.
- Trained a multi-layer 2D Convolutional Neural Network (CNN) in PyTorch to classify music files across 10 discrete genres, achieving high evaluation accuracy.
- Deployed an interactive Streamlit inference dashboard allowing users to upload audio files and visualize real-time frequency distribution charts.

ZIS & Waqf Financial Accounting Management — Lazismu Lengkong *2024*

Laravel, Livewire, Tailwind CSS, MySQL, DomPDF | Repo: github.com/BroAegg/Lazismu_Lengkong-php

- Engineered a sharia-compliant financial donation portal adhering to PSAK 109 accounting standards, cleanly segmenting 6 distinct fund clusters and 8 asnaf beneficiary categories.
- Developed dynamic real-time gold-price nisab calculators using Laravel Livewire, comprehensive audit trail logging, and automated PDF invoice generation.

TECHNICAL SKILLS SUMMARY

Languages: TypeScript, JavaScript, Python, PHP, C/C++ (Embedded), SQL, HTML5, CSS3.

Frontend & Mobile: Next.js 14, React.js, Tailwind CSS, Capacitor (Android APK), Inertia.js, Vite, Framer Motion.

Backend & Cloud: Laravel 11, Node.js, Express, PostgreSQL, MySQL, Redis, Supabase, RESTful APIs, WebSockets.

AI, Vision & Hardware: PyTorch, TensorFlow, OpenCV, Tesseract OCR, Librosa DSP, ESP32, RFID (RC522), Bluetooth SPP.

Tools & Practices: Git, GitHub, Turbopack, Linux/POSIX, Vercel, Postman, Headless Chromium, ACID Database Transactions.